

Phase-field fracture models and MSPIN

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Abstract

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1 Introduction

[What is brittle fracture]
[What is phase-field model]
[How has phase-field been computed in the past, with what limitations]
[Recent advances]
[Current proposition]
[Outline of paper]

2 Background

[Mathematical description of phase-field]
The phase-field energy may be expressed as

$$E_\ell(\mathbf{u}, \mathbf{v}) = \int_{\Omega} \text{tensors} d\Omega + \int_{\partial\Omega} \left(\frac{\omega(\mathbf{v})}{4\ell} + \ell |\mathbf{v}|^2 \right) dS. \quad (1)$$

[Irreversibility condition]

2.1 Alternate Minimization

[AltMin, using notation from mathematical description of phase-field]
AltMin may be viewed as a field-split multiplicative Schwarz method. The two ‘fields’ of damage and displacement are treated separately and sequentially. This allows the use of some of the techniques reserved for Schwarz methods to be applied here. In particular, in Section 2.3 we will examine applying MSPIN to the problem.

2.2 Active set methods

2.3 MSPIN

MSPIN has been previously applied to phase-field fracture models in (nb: cite 3K). There, the authors used cubic backtracking to globalize the Newton direction. Note that their model differs slightly from the one presented here.

2.4 Cubic Backtracking

Many applications of Newton’s method require globalization techniques to achieve convergence when starting away from the minimum. When the Newton direction overshoots the minimum, these globalization techniques should search for a minimum between the starting point and the destination, usually through a line search.

Cubic backtracking (nb: cite) performs this line search by repeatedly interpolating cubic polynomials and finding their minima. Note that a cubic polynomial interpolant requires four pieces of information, but Newton’s method only provides three: the value of the function at the starting point, $f(\mathbf{x})$; the value of the function in the Newton direction, $f(\mathbf{x} + \mathbf{p})$, and; the derivative of the function in the Newton direction, $\mathbf{p} \cdot \nabla f(\mathbf{x})$.

To get a fourth piece of information, first form a quadratic interpolant with this information. Find the minimum of this polynomial and test the value of the function there, thus getting enough information for a cubic polynomial. If the function is smaller at this point, then this ends the line search. Otherwise, find the minimum of the cubic interpolant and repeat. This algorithm is summarized in Algorithm 2.4.

Algorithm 1 Cubic backtracking (nb: cite)

1: inputs

3 Parallelogram globalization

When using MSPIN, one has two potential directions for minimization: the AltMin direction $\mathbf{q}$ and the Newton direction $\mathbf{p}$. Let’s try using both while applying similar ideas to cubic backtracking. Rather than find the minimum of a single variable cubic polynomial, find the minimum of a two-variable quadratic polynomial.

For this interpolant, we need six pieces of information. Five are provided by MSPIN: the value of the function at the starting point, $f(\mathbf{x})$; the value of the function in the AltMin direction, $f(\mathbf{x} + \mathbf{q})$; the value of the function in the Newton direction, $f(\mathbf{x} + \mathbf{p})$; the derivative of the function in the AltMin direction, $\mathbf{q} \cdot \nabla f(\mathbf{x})$, and; the derivative of the function in the Newton direction, $\mathbf{p} \cdot \nabla f(\mathbf{x})$.

One choice for a sixth piece of information is the value of the function at a point symmetric to the starting point. Suppose that the AltMin and Newton directions form two sides of a parallelogram. Then the other corners of the parallelogram are the starting point and $\mathbf{x} + \mathbf{p} + \mathbf{q}$. It is at this new point that we evaluate the function and find our sixth piece of information.

We seek a point $\mathbf{x} + \alpha\mathbf{q} + \beta\mathbf{p}$ that minimizes the objective function. The interpolant appears as $P(\alpha, \beta) = a\alpha^2 + b\alpha\beta + c\beta^2 + d\alpha + e\beta + f$. It is straightforward to show how our six pieces of information relate to our six coefficients:

$$\begin{aligned} d &= \mathbf{q} \cdot \nabla f(\mathbf{x}), & e &= \mathbf{p} \cdot \nabla f(\mathbf{x}), & f &= f(\mathbf{x}), \\ a &= f(\mathbf{x} + \mathbf{q}) - d - f, & c &= f(\mathbf{x} + \mathbf{p}) - e - f, \\ b &= f(\mathbf{x} + \mathbf{q} + \mathbf{p}) + f - f(\mathbf{x} + \mathbf{q}) - f(\mathbf{x} + \mathbf{p}). \end{aligned}$$

The minimum of this interpolant occurs at

$$\alpha = \frac{be - 2cd}{4ac - b^2}, \quad \beta = \frac{-db - 2ae}{4ac - b^2}.$$

It's possible for the minimum of the polynomial to occur far outside of the parallelogram. This may be acceptable, but could also destabilize convergence. To stay within the parallelogram, we can also check the minima on the boundary:

$$(0, -e/2c), \quad (-d/2a, 0), \quad (1, -(e+b)/2c), \quad (-(d+b)/2c, 1).$$

In practice, it is advisable to have β , the amount to step in the Newton direction, be between -1 and 1, while α , the amount to step in the AltMin direction, can be any positive value. We will see examples of this behaviour in the next section.

Before accepting the critical point of the interpolant as the next step in the algorithm, one should perform the second derivative test to confirm that this point is in fact a minimum. The determinant of the Jacobian matrix is $D = 4ac - b^2$, used in the denominator of both α and β . If $D > 0$ and $a > 0$, then the critical point is indeed a minimum. Otherwise, this point may be a maximum or a saddle point. This becomes more likely when the calculation of the gradient of the function is subject to numerical error, especially when the magnitude of the gradient is small in the AltMin and Newton directions. To compensate for these situations, one can try replacing the information from the gradient with more evaluations of the function, resulting in a P2 finite element:

$$\begin{aligned} a &= \frac{1}{2} (f(\mathbf{x} + 2\mathbf{q}) + f) - f(\mathbf{x} + \mathbf{q}), & d &= f(\mathbf{x} + \mathbf{q}) - f - a, \\ c &= \frac{1}{2} (f(\mathbf{x} + 2\mathbf{p}) + f) - f(\mathbf{x} + \mathbf{p}), & e &= f(\mathbf{x} + \mathbf{p}) - f - c, \end{aligned}$$

with b and f remaining the same. This requires only two more evaluations of the objective function. If the new interpolant still fails the second derivative test, then the next step in the algorithm can be chosen as that which has the lowest value of the objective function from amongst the five values already found.

4 Numerical comparisons